

Area Courses

Signal and Image Processing

Fundamental Courses

CSCI 455X Introduction to
 Programming Systems Design
 4 units

or

EE 538 Computing Principles for
 Electrical Engineers
 2 units

EE 483 Introduction to Digital Signal
 Processing
 4 units

EE 503 Probability for Electrical and
 Computer Engineers
 4 units

EE 510 Linear Algebra for Engineering
 4 units

Mathematical Foundations

EE 512 Stochastic Processes for
 Financial Engineering
 4 units EE 503, (EE 510 or EE 518)

EE 562 Random Processes in
 Engineering
 4 units EE 503, EE 510

Law and Intellectual Property

EE 682 Law and Intellectual Property
 for Engineers
 3 units
 EE 503

Signal and Image Processing

EE 566 Optical Information Processing
 4 units

EE 575 Computational Differential
 Geometry for Engineers
 3 units

EE 592 Computational Methods for
 Inverse Problems
 4 units EE 483, EE 510
 EE 503

EE 659 Advanced Topics in Interconnection
 Network Design and Analysis
 4 units EE 557
 EE 450, 451, 452, or 455

EE 569 Introduction to Digital Image
 Processing
 4 units
 EE 503

EE 583 Statistical Signal Processing
 3 units EE 503

EE 596 Wavelets and Graphs for Signal
 Processing and Machine Learning
 4 units EE 483, EE 510

Data Analysis and Machine Learning

EE 517 Statistics and Data Analysis for
 Engineers
 4 units EE 503

EE 518 Mathematics and Tools for
 Financial Engineering
 4 units

EE 559 Machine Learning I: Supervised
 Methods
 3 units EE 503, EE 510
 EE 541

EE 563 Estimation Theory
 3 units EE 503

EE 660 Machine Learning II:
 Mathematical Foundations and Methods
 3 units EE 503, EE 510, EE 559
 EE 538, EE 541

EE 669 Multimedia Data Compression
 4 units
 EE 503

EE 689 Computational Intelligence and
 Neural Learning
 4 units
 EE 483, EE 503, EE 510

Speech, Biomedical, Audio, and Other Applications

EE 519 Speech Recognition and
 Processing for Multimedia
 3 units EE 483

EE 522 Immersive Audio Signal
 Processing
 4 units

EE 523 Advanced Biomedical Imaging
 3 units EE 483

EE 586L Advanced DSP Design
 Laboratory
 4 units EE 483

EE 591 Magnetic Resonance Imaging
 and Reconstruction
 3 units EE 483
 EE 503

EE 619 Advanced Topics in Automatic
 Speech Recognition
 3 units
 EE 503, EE 519, CSCI 544

Legend

Grouping

EE 000 Course Title

Course Units

Recommended Prep.

Prerequisite Courses

Corequisite Courses

This chart shows course relationships

Please check the University Catalogue for specific course details including any recommended preparatory courses and Degree Requirements